

Chem 1041 Assignment (20%)

Submission date: Nov. 02, 2022

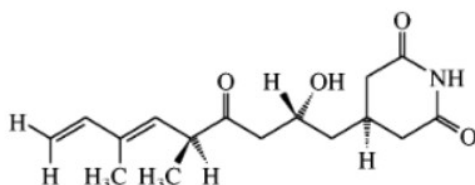
1. a) Explain why the boiling points of alcohols are much higher than those of ethers or hydrocarbons with similar molecular weight as shown below.

	$\text{CH}_3\text{CH}_2\text{OH}$	$\text{CH}_3\text{-O-CH}_3$	$\text{CH}_3\text{CH}_2\text{CH}_3$
Mol. Wt	46	46	44
Boiling point	+78.5 °C	-24 °C	-42 °C

- b) Why carboxylic acids have high boiling points than those alcohol with comparable molecular weight? For example, acetic acid and propyl alcohol, which have the same formula weight (60), boils at 118 °C and 97 °C, respectively.

	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	$\text{CH}_3\overset{\text{O}}{\parallel}\text{COH}$
Boiling point	97 °C	118 °C

2. *Streptimidone* is an antibiotic and has the structure shown. How many diastereomers of streptimidone are possible? How many enantiomers? Using the E, Z and R, S descriptors, specify all essential elements of stereochemistry of streptimidone.



3. An organic compound **A** ($\text{C}_3\text{H}_6\text{O}$) is quite resistant to oxidation. Up on reduction **A** forms another compound **B** ($\text{C}_3\text{H}_8\text{O}$). **B** reacts with HBr to give alkyl bromide **C**. The Grignard reagent derived from **C** reacts with **A** to give **D** ($\text{C}_6\text{H}_{14}\text{O}$). What are the structures of **A**, **B**, **C** and **D**? Write equations for all the reactions.
4. Explain the following with examples.
- Rearrangement reactions
 - Migration to electron deficient carbon- Wagner-Meerwien Rearrangement
 - Migration to electron deficient oxygen-The Bayaer-Villiger Oxidation
 - Migration to electron deficient nitrogen-Beckmann Rearrangement, Hofmann Rearrangement
5. a) What is Diels Alder reaction? Define dienes and dienophiles.
- b) Write the mechanism for Diels Alder reaction. Explain with examples.